

Multi-Agent Reinforcement Learning for Air Handling Unit Control

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Abstract

Air Handling Units (AHUs) are key components of HVAC systems and contribute significantly to building energy consumption. This thesis investigates a multi-agent reinforcement learning framework for coordinated AHU control, where multiple interacting actuators are controlled cooperatively to improve energy efficiency while maintaining comfort, ventilation, and operational constraints.

1 Introduction

Importance:

Buildings account for a significant share of global energy consumption and carbon emissions, with heating, ventilation, and air conditioning (HVAC) systems representing one of the largest end-use loads in both commercial and residential buildings [1, 2]. In the United States, buildings consume nearly 40% of total energy use, and approximately half of this demand is associated with HVAC operation [1]. Therefore, improving HVAC control performance is an important pathway for reducing building energy use and operational cost.

Among HVAC subsystems, Air Handling Units (AHUs) play a central role in regulating airflow, supply air temperature, ventilation rate, and indoor air quality. A typical AHU consists of multiple interacting components, including supply fans, exhaust fans, heating coils, dampers, and rotary heat exchangers. These components jointly influence airflow distribution, thermal comfort, heat recovery, and energy consumption. In practical building management systems, such subsystems are often controlled using multiple independent rule-based or PID loops. However, due to strong thermal and airflow coupling among actuators, independent local control may lead to suboptimal global performance, excessive fan power, unnecessary heating or cooling demand, and unstable operating behavior.

Solution: To overcome these limitations, advanced supervisory control methods have received increasing attention. Model Predictive Control (MPC) has demonstrated strong potential by explicitly incorporating system dynamics, future disturbance forecasts, operational constraints, and multi-objective optimization [3–5]. Nevertheless, its practical effectiveness depends heavily on accurate models and significant engineering effort for modeling, calibration, and maintenance. Recent studies have therefore extended MPC toward data-driven, robust, hierarchical, and smart-grid-oriented frameworks, where HVAC

operation can be jointly optimized with energy storage, renewable generation, electricity pricing, and demand response [6, 7].

Recently, reinforcement learning (RL) has emerged as a promising data-driven alternative that learns control policies directly through interaction with the environment, thereby reducing dependence on explicit first-principles models [1, 2, 8, 9]. Prior studies have shown that RL-based building control can reduce operating cost and energy demand while maintaining thermal comfort. For example, simulation-based DRL studies have reported energy savings of approximately 10%–30% under different building conditions [2], while a practical real-building deployment achieved a 16.7% reduction in heating demand compared with a conventional rule-based controller [8]. More recent studies further address practical barriers such as unsafe exploration, sample inefficiency, and real-world deployment by incorporating expert guidance, prediction models, IoT/BEMS data, and comfort-aware reward design [10, 11].

The coupled structure of AHUs naturally motivates the use of multi-agent control strategies, where different controllers cooperate to optimize overall system performance. Compared with fully centralized control, Multi-Agent Reinforcement Learning (MARL) provides improved scalability and enables decentralized decision-making for complex HVAC systems with multiple interacting actuators. Existing MARL studies mainly focus on zone-level or multi-building coordination problems [12, 13], while relatively limited attention has been given to actuator-level coordination inside AHU systems. Therefore, this thesis proposes a MARL-based actuator-level AHU control framework aiming to improve energy efficiency, thermal comfort, ventilation performance, and operational robustness.

2 Background and Related Work

2.1 Conventional HVAC and MPC-Based Control

2.1.1 Rule-based and PID Control

Conventional HVAC control strategies, including on/off logic, schedule-based control, and PID controllers, are widely used in practice due to their simplicity, robustness, and low implementation cost [3]. In AHU systems, independent local control loops are typically assigned to airflow regulation, temperature tracking, pressure control, valve actuation, and heat recovery operation.

However, buildings exhibit nonlinear and time-varying thermal dynamics influenced by weather conditions, occupancy variation, solar radiation, internal heat gains, and equipment interactions [4]. Rule-based and PID-based controllers are usually designed for fixed operating conditions and often require manual retuning when the building environment changes. As a result, they may operate conservatively and fail to explicitly optimize long-term objectives such as energy efficiency, comfort, ventilation, and actuator coordination.

From a practical perspective, HVAC control also depends on sensing and communication infrastructure. IoT-based HVAC monitoring systems demonstrate that real-time control requires reliable sensors, PLCs, communication modules, cloud platforms, and remote human–machine interfaces [14]. This is important for learning-based AHU control, since reinforcement learning relies on high-quality observations and reliable actuation channels.

2.1.2 MPC Framework

Model Predictive Control (MPC) is widely regarded as one of the most promising advanced control methods for energy-efficient buildings. MPC uses a predictive model of building thermal dynamics to optimize future control inputs over a finite horizon while considering constraints, disturbances, and multi-objective trade-offs [4, 5].

A simplified MPC objective for HVAC control can be written as:

$$J = \sum_{k=0}^N (\alpha E_k + \beta C_k + \lambda U_k),$$

where E_k denotes energy consumption, C_k represents thermal comfort violation, U_k denotes control effort, and α , β , and λ are weighting coefficients. This formulation reflects the typical trade-off between energy saving, comfort maintenance, and smooth control actions.

Compared with rule-based and PID controllers, MPC can exploit building thermal mass, perform pre-heating or pre-cooling, shift loads away from peak-price periods, and anticipate future disturbances such as outdoor temperature and occupancy changes [4]. However, MPC deployment remains limited by model development effort, uncertainty handling, communication integration, computational complexity, and the need for expert tuning [5].

2.1.3 Hierarchical and Smart-Grid MPC

Recent MPC research has moved beyond single-zone temperature tracking toward hierarchical and smart-grid-oriented building energy management. Data-driven MPC frameworks have been proposed to coordinate HVAC operation with photovoltaic generation, energy storage, base loads, and grid power exchange under uncertainty [?]. These studies show that building control problems are often multi-time-scale: HVAC systems respond slowly due to thermal inertia, while energy storage systems and grid interactions can be adjusted more frequently.

Hierarchical MPC provides another important direction. In such frameworks, a high-level controller schedules long-horizon energy management decisions, while a low-level controller tracks reference signals and handles short-term uncertainty [7]. This decomposition is conceptually relevant to AHU control, because AHU operation can also be separated into interacting control tasks, such as airflow regulation, supply air temperature control, heat recovery control, and ventilation management.

These studies suggest that complex HVAC systems are difficult to control using a single monolithic controller. Instead, decomposition into layers, subsystems, or agents can improve scalability and reduce control complexity. This idea provides a useful foundation for MARL-based AHU control.

2.2 Reinforcement Learning for HVAC Systems

2.2.1 MDP Formulation

Reinforcement Learning (RL) has gained increasing attention as an alternative to model-based HVAC control. Unlike MPC, RL does not require an explicit mathematical model and instead learns a control policy through repeated interaction with the environment [2, 9].

RL-based HVAC control is commonly formulated as a Markov Decision Process (MDP):

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma),$$

where $\mathcal{S}$ denotes the state space, $\mathcal{A}$ denotes the action space, P represents the transition dynamics, R is the reward function, and γ is the discount factor.

The objective is to learn a policy π that maximizes the expected cumulative reward:

$$\max_{\pi} E \left[\sum_{t=0}^{\infty} \gamma^t r_t \right].$$

In HVAC applications, states may include indoor temperature, airflow rate, weather conditions, occupancy information, energy consumption, and historical measurements. Actions may correspond to temperature setpoints, fan speeds, damper positions, valve openings, or supply air temperature commands. The reward function usually balances energy consumption and comfort satisfaction.

2.2.2 Deep RL Algorithms

With the development of deep neural networks, Deep Reinforcement Learning (DRL) has enabled RL methods to handle high-dimensional and nonlinear HVAC control problems [1]. Early studies applied Deep Q-Networks (DQN) to building HVAC control in EnergyPlus simulation environments, where DRL was shown to reduce energy cost while maintaining indoor temperature within comfort bounds [1].

Later studies adopted actor-critic algorithms such as A3C, DDPG, PPO, TD3, and SAC to support continuous control and improve training stability [2, 9]. A practical whole-building DRL framework using calibrated building energy models achieved a 16.7% reduction in heating demand with more than 95% probability relative to a rule-based baseline [8]. Expert-guided DRL has also been proposed to reduce unstable early exploration; for example, one study reported approximately 15% energy reduction, 61% reduction in temperature violations, and 23% total cost reduction compared with rule-based control [10].

Recent prediction-guided DRL studies further combine time-series forecasting models with RL agents. A GRU-PPO framework using real IoT/BEMS data directly controlled AHU fan speed and pressure-independent control valve openings, achieving up to 11.85% energy savings while maintaining PMV within the ANSI/ASHRAE 55 comfort range of $[-0.5, 0.5]$ [11]. This is especially relevant to this thesis because it moves beyond zone temperature setpoint control toward actuator-level AHU optimization.

2.2.3 Simulation vs Real Deployment

Although DRL has demonstrated strong potential, most existing studies remain simulation-driven [2, 9]. Simulation platforms such as EnergyPlus and TRNSYS provide safe and flexible environments for offline training, but policies trained in simulation may not transfer well to real buildings due to model mismatch, unmodeled disturbances, and incomplete observations.

Several practical barriers remain, including sample inefficiency, unsafe exploration, reward sensitivity, transferability across buildings, and sim-to-real gaps [9]. Recent studies therefore increasingly incorporate expert knowledge, prediction models, event-triggered

logic, IoT infrastructure, and real operational data to improve practical deployment feasibility [10, 11, 14, 15].

For AHU control, deployment challenges are particularly important because unsafe exploration may cause discomfort, poor ventilation, excessive fan energy, or unstable actuator behavior. Therefore, an effective RL-based AHU controller should combine data-driven learning with operational constraints and reliable sensing infrastructure.

2.3 Multi-Agent Reinforcement Learning

2.3.1 MARL Architecture

As HVAC systems become increasingly interconnected and high-dimensional, centralized RL approaches face challenges in scalability, coordination, and training stability. Multi-Agent Reinforcement Learning (MARL) extends conventional RL by introducing multiple cooperative agents operating in a shared environment.

In a MARL setting, the global action can be decomposed into multiple agent-level actions:

$$\mathcal{A} = \{a_1, a_2, \dots, a_n\},$$

where each a_i is selected by an individual agent controlling a subsystem or actuator. The global reward may be designed as:

$$r_t = -\alpha E_t - \beta C_t - \lambda V_t,$$

where E_t denotes energy consumption, C_t denotes comfort violation, and V_t denotes ventilation or operational constraint violation.

Many MARL frameworks adopt centralized training with decentralized execution (CTDE). During training, agents may access shared information to improve coordination; during execution, each agent acts based on local observations. This architecture is suitable for practical building automation systems because it combines coordinated learning with decentralized implementation.

2.3.2 Multi-Zone and Multi-Building MARL

Recent MARL studies have investigated coordinated control in multi-zone and multi-building HVAC systems [12, 13]. In multi-zone buildings, agents may coordinate zone-level energy allocation and thermal comfort. In multi-building systems, agents may coordinate HVAC operation, energy storage, photovoltaic utilization, and grid-level voltage constraints.

For example, MARL has been applied to zone-energy-allocation optimization in multi-zone HVAC systems and to grid-aware coordination control of building clusters [12, 13]. These studies demonstrate that cooperative learning can improve scalability and control performance compared with fully centralized approaches.

However, MARL also introduces additional challenges, including non-stationarity, credit assignment, communication design, and convergence stability [9]. These challenges are particularly relevant when multiple controllers affect the same thermal and airflow environment.

2.3.3 Actuator-Level AHU Coordination

Most existing MARL studies focus on zone-level or building-level coordination, while relatively limited attention has been given to actuator-level coordination inside AHU systems. This thesis focuses on this gap.

A practical AHU topology contains multiple coupled actuators, such as supply fans, exhaust fans, rotary heat exchangers, heating valves, heating coils, and dampers. These components jointly determine supply airflow, supply air temperature, ventilation effectiveness, heat recovery performance, and energy consumption. Therefore, controlling them independently may produce conflicting or inefficient behaviors.

From a MARL perspective, each actuator can be associated with an individual agent. For example, one agent may regulate supply fan speed, another may control exhaust fan speed, another may adjust the heating valve, and another may control the rotary heat exchanger. These agents share the same physical environment and must coordinate their actions to satisfy global objectives.

This actuator-level formulation is different from conventional multi-zone MARL. Instead of assigning one agent to each thermal zone or building, the proposed direction assigns agents to interacting AHU components. This structure is well aligned with real AHU topology and practical building management systems, where multiple local controllers already exist but lack coordinated learning capability.

3 Problem Formulation

The AHU control problem can be formulated as a sequential decision-making problem.

4 Methodology

This work adopts a multi-agent reinforcement learning framework.

The model will be trained using simulation data or historical operational data.

5 Results

The proposed method is expected to:

6 Conclusion

This draft outlines a reinforcement learning-based approach for AHU control using a multi-agent framework. Future work will focus on implementation, training, and evaluation in realistic simulation environments.

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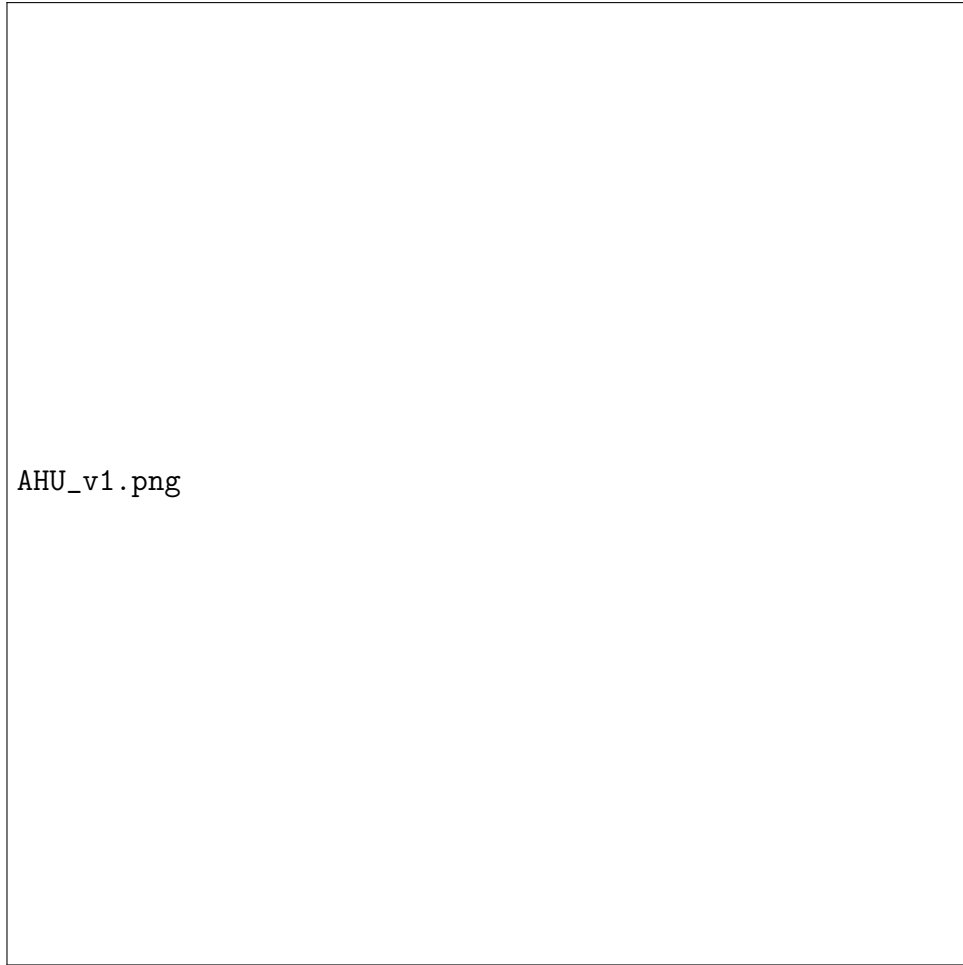


Figure 1: Architecture of the AHU system

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